MANVI SANJAY MESHRAM

Pune, Maharashtra, India | +91 9145344474 | manvimeshram1508@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Final-year Computer Engineering student and Certified Data Scientist (ExcelR) with hands-on experience building end-to-end ML pipelines — from preprocessing and feature engineering through model training, evaluation, and deployment-ready outputs. Proficient in Python, SQL, and the core ML/DS stack (Scikit-learn, Pandas, NumPy, OpenCV), with project work spanning healthcare AI, predictive analytics, and safety analytics. Seeking a Data Science / Machine Learning internship to apply this foundation in a production engineering environment.

TECHNICAL SKILLS

Languages & Querying: Python, SQL

ML & Modeling: Supervised Learning, Predictive Modeling, Model Evaluation & Hyperparameter Tuning, Cross-Validation, Feature Engineering, Exploratory Data Analysis (EDA)

Libraries & Frameworks: Scikit-learn, Pandas, NumPy, OpenCV, Matplotlib, Seaborn

Data Pipelines & Tools: Jupyter Notebook, Power BI, Excel, Statistical Analysis

Applied Domains: Healthcare Analytics, Medical Image Analysis, Computer Vision, Signal Processing, Safety/Risk Analytics

PROJECTS

Image Processing for Cancer Detection — Healthcare AI | Python, Scikit-learn, OpenCV Jan 2024 – Apr 2024

- Built an end-to-end ML pipeline for automated cancer detection from medical images — covering image preprocessing, feature extraction, and multi-class classification.
- Applied image segmentation and noise-reduction techniques (thresholding, morphological operations) using OpenCV to improve diagnostic image quality.
- Trained and benchmarked multiple classifiers (SVM, Random Forest, KNN), tuning hyperparameters and applying cross-validation to optimize performance.
- Prioritized recall over raw accuracy to minimize false negatives — a critical constraint in cancer screening workflows — and reported model performance with precision/recall trade-off analysis.

Heart ECG Signal Analysis & Anomaly Detection — Predictive Analytics | Python, Scikit-learn Aug 2023 – Dec 2023

- Developed a predictive model for ECG time-series classification to flag cardiac anomalies, applying signal processing techniques for noise filtering and feature extraction.
- Implemented and compared supervised models (Logistic Regression, Decision Tree, Random Forest) for binary and multi-class arrhythmia detection.
- Evaluated models using AUC-ROC, precision, and recall to validate clinical reliability of predictions.
- Ran structured EDA on ECG datasets to surface patterns correlating with cardiac conditions, directly informing feature selection and model design.

Road Accident Analysis System — Safety Analytics | Python, Pandas, Matplotlib, Power BI Apr 2023 – Jul 2023

- Ran end-to-end EDA on large-scale accident datasets to identify high-risk zones, peak accident windows, and contributing factors (weather, speed, road type).
- Engineered features from raw accident records and applied statistical testing to validate hypotheses on risk-factor correlations.
- Built interactive Power BI dashboards delivering geo-spatial and trend insights to support data-driven road-safety recommendations.
- Used Python (Pandas, Matplotlib, Seaborn) for data cleaning, aggregation, and publication-ready visual storytelling.

EDUCATION

B.Tech in Computer Engineering Expected May 2026

Dr. Babasaheb Ambedkar Technological University, Lonere, Maharashtra

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Probability & Statistics, Digital Image Processing, Artificial Intelligence, Computer Networks

CERTIFICATIONS & TRAINING

Professional Data Science Certification — ExcelR Solutions	2023
Machine Learning & Data Analytics Training — ExcelR Solutions	2023
Python for Data Science — ExcelR Solutions	2023

LANGUAGES

English (Professional) • Hindi (Native) • Marathi (Native)